A Multimodal Network Optimization Framework for Cost–Time Efficient Logistics Planning

Ajeenckya Mahadik
avmahadik@wisc.edu
University of Wisconsin–Madison

***Abstract*—Modern logistics and field-sales operations frequently require route planning across large geographic regions while balancing competing objectives: minimizing monetary cost and minimizing travel time. This work presents *OptiRoute*, a constrained multimodal route optimization framework that integrates flights, trains, and buses into a unified directed graph and computes optimal origin–destination travel paths under operational constraints. For each target destination, we solve a binary integer linear program (BILP) that enforces flow conservation, restricts the number of legs (stopovers), and incorporates mode feasibility constraints for remote destinations. Using U.S.-wide transportation datasets and product-level revenue data, the system optimizes 2,040 routes from a central hub (Chicago) to revenue-maximizing state capitals. Results demonstrate that multimodal optimization can achieve broad reachability while keeping transportation expenditure negligible relative to revenue, and provide interpretable, constraint-compliant itineraries suitable for decision support.**

***Index Terms*—Multimodal transportation, shortest path, integer linear programming, constrained routing, logistics optimization, network flow.**

I. Introduction

Logistics planning is fundamentally about making trade-offs. Flights are fast but costly, buses are cheap but slow, and trains often occupy the middle ground. In real deployments, committing to a single mode can lead to suboptimal decisions—either overspending on air travel or wasting operational time on ground travel. The core contribution of OptiRoute is a unified *multimodal* optimization layer that selects routes across flights, trains, and buses while respecting operational constraints like stop limits and mode feasibility.

Unlike the classical Traveling Salesman Problem (TSP), which seeks a minimum-cost tour visiting many cities exactly once, the operational setting here is naturally decomposed into multiple origin–destination (OD) problems: each product (or assignment) induces a destination city that must be reached from a fixed hub. This enables solving many smaller but still realistic constrained routing problems at scale.

This paper formalizes the multimodal routing problem as a constrained shortest path optimization and demonstrates an exact approach using binary integer linear programming (BILP), following the project logic documented in the provided report. *All details are based on the project document* :contentReference[oaicite:1]index=1.

II. Problem Setting and Data

A. Operational Scenario

We consider an organization headquartered at a hub city (Chicago). For each product instance, revenue data identifies the most valuable target state; the target destination is the *capital city* of that state. For each destination, we compute a route from Chicago to that capital using one or more transportation modes (bus, train, flight). This yields a set of optimized itineraries suitable for sales planning or logistics dispatching.

B. Datasets and Network Construction

The system consumes:

- **Sales/Revenue Data**: product-by-state revenue, used to select revenue-maximizing destinations per product.
- **Salesman Data**: efficiency and salary metadata used for post-optimization assignment reporting (profit calculation and ranking).
- **Transportation Data**: flight, train, and bus route tables with origin, destination, stop lists (if any), prices, and durations.

C. Graph Representation

We construct a directed graph:

$$G = (V, E), \tag{1}$$

where V is the set of cities (hub + capitals + intermediate cities from stop lists), and E is the set of directed transportation legs. Each edge $e \in E$ has:

$$c_e \geq 0 \quad \text{(monetary cost)}, \qquad t_e \geq 0 \quad \text{(travel time)}. \tag{2}$$

Edges also carry a categorical mode label:

$$m_e \in \{\text{bus}, \text{train}, \text{flight}\}. \tag{3}$$

D. Why Data Augmentation / Preprocessing Matters

In multimodal datasets, routes may appear as multi-stop itineraries encoded in a single record (e.g., a bus route with intermediate stops). For optimization, such records must be converted into *atomic legs* so that the solver can choose partial itineraries and mix modes. Therefore, each multi-stop record is decomposed into a sequence of edges:

$$(\text{Chicago} \to s_1), (s_1 \to s_2), \ldots, (s_k \to \text{Destination}).$$

The cost and duration are allocated to legs using the dataset's implied structure (e.g., evenly or proportionally). This decomposition increases connectivity (more feasible paths), reduces modeling ambiguity, and allows meaningful multimodal combinations.

Additionally, infeasible edges (e.g., land routes to geographically isolated destinations) are removed by feasibility rules (Section III-F).

III. Mathematical Formulation

A. Constrained Multimodal Shortest Path

For each destination $d \in V$, we seek a path from source s (Chicago) to d minimizing a composite objective of cost and time, under constraints (stopovers, feasibility).

This can be viewed as a constrained shortest path problem. While classical Dijkstra's algorithm solves the unconstrained single-objective shortest path in polynomial time, additional constraints such as bounded hops/legs make the problem harder; the BILP approach provides a clean mechanism to express and enforce such constraints exactly.

B. Decision Variables

For each directed edge $e \in E$, define a binary decision variable:

$$x_e = \begin{cases} 1, & \text{if edge } e \text{ is used in the route,} \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

C. Objective Function

We minimize a weighted sum of monetary cost and travel time:

$$\min_x \sum_{e \in E} (c_e + \lambda\, t_e)\, x_e. \tag{5}$$

Here, $\lambda \geq 0$ converts minutes into "dollar-equivalent" units. In the original implementation, the effective choice is $\lambda = 1$ (treating 1 minute as equal to \$1 in the objective), which produces cost-conscious routes and makes the trade-off explicit and tunable.

D. Flow Conservation Constraints

Let $\delta^+(v)$ be edges leaving node v and $\delta^-(v)$ edges entering node v. We enforce a unit flow from s to d:

$$\sum_{e \in \delta^+(v)} x_e - \sum_{e \in \delta^-(v)} x_e = \begin{cases} 1, & v = s, \\ -1, & v = d, \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

This ensures that chosen edges form a continuous OD route (no disconnected fragments).

E. Stopover / Leg Bound Constraint

To avoid impractical solutions with excessive transfers, we bound the total number of selected legs:

$$\sum_{e \in E} x_e \leq L, \tag{7}$$

where L is a configurable maximum (e.g., $L = 7$). This constraint is operationally meaningful (limits fatigue, scheduling complexity) and also reduces search space.

F. Mode Feasibility Constraints

Certain destinations are only reachable by specific modes (e.g., islands or isolated regions). Let $\mathcal{M}(d)$ denote allowed modes for destination d. Then:

$$x_e = 0 \quad \forall e \in E \text{ with } m_e \notin \mathcal{M}(d), \tag{8}$$

and similarly, we may prune intermediate-node incidence that would imply infeasible mode transitions into disallowed geography.

G. Binary Domain

$$x_e \in \{0, 1\}, \quad \forall e \in E. \tag{9}$$

H. Interpretation and Guarantees

Because (5)–(8) define a linear objective with linear constraints and binary variables, this is a Mixed-Integer Linear Program (MILP). Solving it yields a globally optimal route under the stated constraints. This provides a strong academic advantage: *results are not heuristic*; they are exact with respect to the model.

IV. Solution Approach

A. Per-Destination Decomposition

Rather than solving one monolithic problem for all products and destinations, we solve a separate BILP for each destination city d. This decomposition is valid because routing costs do not couple across products in the current scope (no shared capacity constraints). The advantage is scalability: many small MILPs solve quickly and independently.

B. Caching and Reuse

Many products may map to the same destination capital. OptiRoute caches computed routes (path + cost + time) per destination, reusing solutions to avoid redundant solves. This dramatically reduces runtime when destinations repeat.

C. Algorithm Outline

V. Evaluation Metrics and Reporting

A. Route Quality Metrics

For each computed route, we report:

- **Transport Cost:** $C(d) = \sum_{e \in E} c_e x_e$
- **Travel Time:** $T(d) = \sum_{e \in E} t_e x_e$
- **Composite Objective:** $J(d) = \sum_{e \in E} (c_e + \lambda t_e) x_e$
- **Mode Sequence:** inferred from the selected edges
- **Leg Count:** $\sum_{e \in E} x_e$

B. Business Metrics (Post-Optimization)

Although the routing optimization is not directly profit-maximizing, it supports decision-making via a net-gain analysis:

$$\text{NetGain} = \text{Revenue} - \text{Salary} - \text{TransportCost}. \tag{10}$$

This yields interpretable financial outcomes per assignment.

VI. Results

Algorithm 1 OptiRoute (Per-Destination Multimodal Routing)

Input: Hub city s, destinations $\mathcal{D}$, transport datasets, max legs L, weight λ
Build graph $G = (V, E)$ by decomposing multi-stop records into legs
Prune infeasible edges (mode/geography rules)
Initialize route cache $\mathcal{C} \leftarrow \emptyset$
for each destination $d \in \mathcal{D}$ **do**
  if $d \in \mathcal{C}$ **then**
    Retrieve cached solution
  else
    Build BILP with constraints (6)–(7) and objective (5)
    Solve using MILP solver (e.g., GLPK/CBC)
    Store optimal route in $\mathcal{C}[d]$
  end if
end for
Output: Optimal paths for all destinations, with cost/time/mode breakdown

A. Aggregate Outcomes

Across all companies and products, OptiRoute optimized 2,040 routes (51 companies $\times$ 40 products). The system produced routes with extremely low transportation expenditures relative to revenues.

TABLE I
AGGREGATE RESULTS (REPORTED)

Metric	Value
Total Routes Optimized	2,040
Total Revenue	\$171.62M
Total Salary	\$123.91M
Total Transportation Cost	\$85.41K
Overall Net Gain	\$47.63M

B. Interpretation

Two consistent phenomena emerge:

1) **Ground modes dominate in the contiguous U.S.** Under $\lambda = 1$, the solver frequently prefers buses and trains due to their substantially lower monetary cost, even when total travel time increases moderately. This behavior highlights the strong influence of ticket price asymmetries across modes and confirms that, for most mainland destinations, cost savings outweigh moderate time penalties in the composite objective.
2) **Flights are selected when required by feasibility.** For geographically isolated destinations (e.g., Hawaii and Alaska), mode feasibility constraints eliminate ground-based alternatives, forcing air travel. In these cases, the optimizer consistently selects the lowest-cost feasible flight itinerary, demonstrating correct constraint enforcement and robust handling of disconnected subgraphs.

C. Visualization

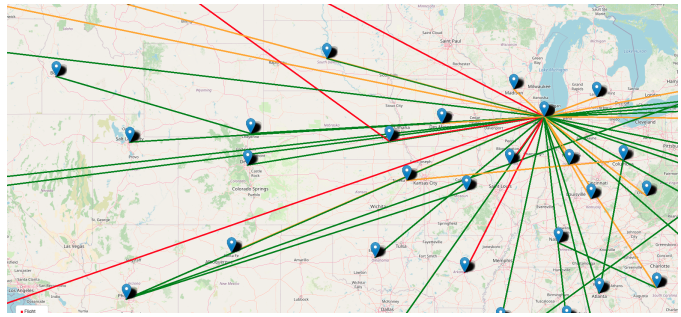

Fig. 1. Multimodal route visualization from hub to destinations. Color by mode.

VII. Discussion

A. Modeling Choices and Their Consequences

The composite objective (5) is intentionally simple and operational. However, it implies a specific trade-off: treating one minute as worth λ dollars. If λ is small, the optimizer prioritizes cost savings (more buses/trains). If λ is large, the optimizer prioritizes speed (more flights).

A key insight is that *objective design is policy*. The same network and constraints can yield drastically different routing strategies depending on how decision-makers value time.

B. Why MILP Instead of Dijkstra?

Dijkstra solves unconstrained shortest path efficiently but does not naturally incorporate the stopover constraint (7) or mode feasibility constraints (8) without state augmentation. The MILP approach provides:

- Direct, exact enforcement of operational constraints
- Interpretability via selected edges and dual reasoning
- Flexibility to add future constraints (e.g., max time, max budget, mandatory mode)

C. Limitations

- **Time is not monetized via wages.** Salary is accounted as a fixed cost per assignment, but longer routes do not increase labor cost. A more realistic model would incorporate an hourly wage multiplier.
- **Static network.** Real-world schedules, delays, and dynamic pricing are not modeled.
- **Independent OD routing.** Capacity coupling and multi-stop tour planning (VRP/TSP variants) are outside current scope.

VIII. Extensions and Future Work

A. Bi-objective (Pareto) Optimization

Rather than collapsing cost and time into one scalar objective, a research extension is to compute Pareto-optimal routes:

$$\min(C(d), T(d)) \quad \text{subject to constraints.}$$

This can be done via ϵ-constraint methods (e.g., minimize cost subject to time $\leq \tau$).

B. Labor-Aware Objective

A more realistic objective prices time using the assigned salesman's wage w:

$$\min \sum_{e \in E} (c_e + w \cdot t_e)\, x_e,$$

aligning routing decisions with true operational cost.

C. Multi-Destination Planning (VRP)

If each salesman must visit multiple destinations in a trip window, the problem becomes a vehicle routing problem (VRP) with multimodal legs and time windows. This is publishable-grade work and a natural next step.

IX. Conclusion

This project demonstrates that constrained multimodal routing can be modeled cleanly as a network-flow MILP, producing exact, interpretable itineraries at scale. By integrating transportation data across modes with revenue-driven destination selection, the framework supports logistics and sales planning decisions that are both operationally feasible and economically meaningful. The approach is extensible to multi-objective optimization, labor-aware costing, and full VRP formulations suitable for deployment-oriented research.

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